

Problem

A transistor in its common-emitter mode is specified by

$$[h] = \begin{bmatrix} 200\ \Omega & 0 \\ 100 & 10^{-6}\ \text{S} \end{bmatrix}$$

Two such identical transistors are connected in cascade to form a two-stage amplifier used at audio frequencies. If the amplifier is terminated by a 4-k Ω resistor, calculate the overall A_v and Z_{in} .

Step-by-step solution

Step 1 of 7

Consider the following h parameters for a transistor in its common-emitter mode.

$$[h] = \begin{bmatrix} 200\ \Omega & 0 \\ 100 & 10^{-6}\ \text{S} \end{bmatrix}$$

Calculate the determinant of h parameters.

$$\begin{aligned} \Delta_h &= h_{11}h_{22} - h_{12}h_{21} \\ &= (200\ \Omega)(10^{-6}\ \text{S}) - (0)(100) \\ &= 0.0002 \end{aligned}$$

Consider that the two identical transistors of same h parameters h_a and h_b are cascaded in a two-stage amplifier circuit.

Step 2 of 7

The T parameters are multiplied in the correct order in the cascade connection.

Convert the h parameters to T parameters.

$$\begin{aligned} [T] &= \begin{bmatrix} A & B \\ C & D \end{bmatrix} \\ &= \begin{bmatrix} -\frac{\Delta_h}{h_{21}} & -\frac{h_{11}}{h_{21}} \\ -\frac{h_{22}}{h_{21}} & -\frac{1}{h_{21}} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{0.0002}{100} & -\frac{200\ \Omega}{100} \\ -\frac{10^{-6}\ \text{S}}{100} & -\frac{1}{100} \end{bmatrix} \\ &= \begin{bmatrix} -2 \times 10^{-6} & -2\ \Omega \\ -10^{-8}\ \text{S} & -0.01 \end{bmatrix} \end{aligned}$$

Step 3 of 7

The following are the two T parameters for the two identical stages.

$$[T_a] = \begin{bmatrix} -2 \times 10^{-6} & -2 \Omega \\ -10^{-8} \text{ S} & -0.01 \end{bmatrix}$$

$$[T_b] = \begin{bmatrix} -2 \times 10^{-6} & -2 \Omega \\ -10^{-8} \text{ S} & -0.01 \end{bmatrix}$$

Multiply the T parameters of two identical stages to get a single T parameter for the cascaded stage.

$$\begin{aligned} [T] &= [T_a][T_b] \\ &= \begin{bmatrix} -2 \times 10^{-6} & -2 \Omega \\ -10^{-8} \text{ S} & -0.01 \end{bmatrix} \begin{bmatrix} -2 \times 10^{-6} & -2 \Omega \\ -10^{-8} \text{ S} & -0.01 \end{bmatrix} \\ &= \begin{bmatrix} (4 \times 10^{-12}) + (2 \times 10^{-8}) & (4 \times 10^{-6}) + 0.02 \\ (2 \times 10^{-14}) + (10^{-10}) & (2 \times 10^{-8}) + (10^{-4}) \end{bmatrix} \\ &= \begin{bmatrix} 20.004 \times 10^{-9} & 20.004 \times 10^{-3} \\ 100.02 \times 10^{-12} & 100.02 \times 10^{-6} \end{bmatrix} \end{aligned}$$

Step 4 of 7

Calculate the determinant of T parameters of the two-stage amplifier.

$$\begin{aligned} \Delta_T &= \mathbf{AD} - \mathbf{BC} \\ &= (20.004 \times 10^{-9})(100.02 \times 10^{-6}) - (20.004 \times 10^{-3})(100.02 \times 10^{-12}) \\ &= (2 \times 10^{-12}) - (2 \times 10^{-12}) \\ &= 0 \end{aligned}$$

Convert the T parameters to h parameters of two-stage amplifier.

$$\begin{aligned} [h] &= \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\mathbf{B}}{\mathbf{D}} & \frac{\Delta_T}{\mathbf{D}} \\ -\frac{1}{\mathbf{D}} & \frac{\mathbf{C}}{\mathbf{D}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{20.04 \times 10^{-3}}{100.02 \times 10^{-6}} & \frac{0}{100.02 \times 10^{-6}} \\ \frac{-1}{100.02 \times 10^{-6}} & \frac{100.02 \times 10^{-12}}{100.02 \times 10^{-6}} \end{bmatrix} \\ &= \begin{bmatrix} 200 & 0 \\ -10^4 & 10^{-6} \end{bmatrix} \end{aligned}$$

Step 5 of 7

Consider the load resistance, $R_L = 4 \text{ k}\Omega$.

The following is the two-stage amplifier with the h parameters:

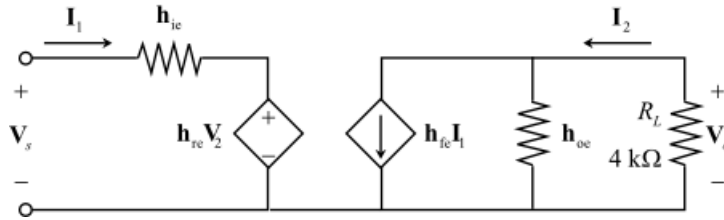


Figure 1

The following are the h parameters for the two-stage amplifier.

Short-circuit input impedance, $h_{ie} = 200 \text{ }\Omega$,

Open-circuit reverse voltage gain, $h_{re} = 0$,

Short-circuit forward current gain, $h_{fe} = -10^4$, and

Open-circuit output admittance, $h_{oe} = 10^{-6} \text{ S}$.

Step 6 of 7

Calculate the overall voltage gain, A_v using the following expression.

$$A_v = \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe} - h_{re}h_{fe})R_L}$$

Substitute $4 \text{ k}\Omega$ for R_L and the values for h parameters.

$$\begin{aligned} A_v &= \frac{-(-10^4)(4 \times 10^3 \text{ }\Omega)}{200 + [(200)(10^{-6}) - (0)(-10^4)](4 \times 10^3 \text{ }\Omega)} \\ &= \frac{4 \times 10^7}{200 + 0.8} \\ &\cong 2 \times 10^5 \end{aligned}$$

Thus, the voltage gain, A_v is 2×10^5 .

Step 7 of 7

Calculate the input impedance, Z_{in} using the following expression.

$$Z_{in} = h_{ie} - \frac{h_{re}h_{fe}R_L}{1 + h_{oe}R_L}$$

Substitute $4 \text{ k}\Omega$ for R_L and the values for h parameters.

$$\begin{aligned} Z_{in} &= 200 - \frac{(0)(-10^4)(4 \times 10^3 \text{ }\Omega)}{1 + (10^{-6})(4 \times 10^3 \text{ }\Omega)} \\ &= 200 - 0 \\ &= 200 \text{ }\Omega \end{aligned}$$

Thus, the input impedance, Z_{in} is $200 \text{ }\Omega$.